

Borhan Javadian

Istanbul, Türkiye | borhan.javadian@sabanciuniv.edu | +90 534 642 7876
github.com/Borhanxj | linkedin.com/in/borhan-javadian | borhanxj.github.io

Research Interests

Computer Vision, Representation Learning, Domain Generalization, Generalized Category Discovery, High-Performance Computing, Parallel Algorithms, GPU Computing, and Sparse Linear Algebra.

Education

Sabancı University

B.Sc. in Computer Science and Engineering

Istanbul, Türkiye

2023 – Expected Jun. 2027

- CGPA: **3.82/4.00**.
- Dean's Honor or High Honor List in every completed semester.
- Academic Success Scholarship (**50% tuition**), 2025–26 and 2026–27; upgraded from a 25% admission scholarship.
- Relevant coursework: Algorithms, Data Structures, Parallel Computing, Computer Networks, Cryptography, Computer and Network Security, Cloud Computing, Operating Systems, and Digital Image and Video Analysis.

Minor in Mathematics

Sabancı University

GPA: **4.00/4.00**

- Coursework: Linear Algebra, Probability, Discrete Mathematics, Differential Equations, Abstract Algebra, Elementary Number Theory, and Mathematical Analysis.
- Earned **A** in every completed course in the minor.

Research Experience

Domain Generalization for Generalized Category Discovery in Remote Sensing

Graduation Project, Sabancı University

Fall 2026 – Present

Prof. Erhan Aptoula

- Working on generalized category discovery when test images come from domains not seen during training, with remote sensing as the main application.
- Reproducing recent GCD and DG-GCD methods and comparing them under the same datasets and evaluation settings.
- Implemented a GCD pipeline with DINO features, k-means, semi-supervised k-means, contrastive training, Hungarian matching, and Old/New/All accuracy.
- Currently studying DG2CD-Net, HiDISC, and related work before moving to the new method in the second part of the project.

Sparse Matrix–Vector Multiplication

Research Project, Sabancı University

Summer 2026 – Present

with guidance from Prof. Kamer Kaya

- Working on different implementations and optimization methods for sparse matrix–vector multiplication.
- Implemented sequential CSR, OpenMP, CUDA, MPI, and hybrid CPU–GPU versions of SpMV.
- Tested matrix reorderings including RCM, AMD, GPS, and graph-based methods and compared their effect on performance.
- Built benchmarking code for comparing methods across different sparse matrices, thread counts, and execution settings.
- More recent work has focused on implementing and comparing newer SpMV methods in a common benchmarking environment.

Fully Homomorphic Encryption for Matrix Multiplication

PURE Research Intern, later Research Assistant, Sabancı University

Jul. 2025 – Feb. 2026

Prof. Erkan Savaş

- Worked on matrix multiplication under the CKKS scheme using Microsoft SEAL.
- Implemented and compared Halevi–Shoup and bicyclic encoding methods in C++.
- Built benchmarks for different matrix sizes and measured runtime, ciphertext rotations, and multiplicative depth.
- Tested batching and chunking methods for encrypted matrix–vector multiplication.
- Presented the work at the Sabancı University PURE research symposium.

Selected Projects

- Generalized Category Discovery

GitHub

- PyTorch implementation of GCD on CIFAR-10 using DINO features, contrastive losses, k-means, semi-supervised k-means, and clustering evaluation.
- SpMV Optimization

GitHub

- C++ SpMV implementation with sequential and OpenMP kernels, blocked methods, matrix reorderings, benchmarks, and result plots.
- Contree-Parallel

- Parallelized optimal classification-tree construction using branch-and-bound, with CPU and GPU implementations.
- FHE Matrix Multiplication

GitHub

- C++ implementation and benchmarks for CKKS matrix operations using Microsoft SEAL.
- ECC Signal Protocol

GitHub

- Implemented a Signal-style messaging protocol using elliptic-curve key exchange, Ed25519 signatures, and forward secrecy.

Teaching Experience

- Learning Assistant

2024 – Present

Sabanci University

- Database Systems (CS 306), Fall 2026.
 - Blockchain: Security and Applications (CS 438), Spring 2025.
 - Logic and Digital System Design (CS 303), Fall 2024–Spring 2025.
 - Calculus I (MATH 101), Spring 2024.
 - Helped with labs, office hours, recitations, grading, and exams.

Industry Experience

- Proteo Cyber Defence AB

Remote

Cybersecurity Intern

Summer 2026

- Worked on attack-surface analysis, vulnerability assessment, network reconnaissance, EDR/XDR, incident response, and detection engineering.

Technical Skills

Programming	Python, C++, C, Go, CUDA, JavaScript, SQL, Bash, LaTeX
Machine Learning	PyTorch, torchvision, scikit-learn, DINO/DINOv2, Vision Transformers, contrastive learning, clustering
Parallel Computing	CUDA, OpenMP, MPI, SLURM, GPU programming, distributed-memory programming, sparse linear algebra
Cryptography	CKKS, Microsoft SEAL, elliptic-curve cryptography
Systems & Cloud	Linux, Git, GitHub, computer networks, AWS, PostgreSQL

Awards and Honors

- Ranked first in Computer and Network Security.
- Highest class score in Parallel Computing, Blockchain: Security and Applications, and Elementary Number Theory.
- Qualified for the Advanced Group, inzva Algorithm Competition Summer Camp, 2026.
- 1st Place and Gold Medal, Caribou Mathematics Contest, 2021.
- Distinction, Fermat Mathematics Contest, University of Waterloo, 2021.

Languages

Persian (Native) | English (C1) | Turkish (Full Professional Proficiency)